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Aerial Robotics Project Report

Pick-and-Place with a Fully Actuated Hexarotor and a 3-DoF Manipulator in Gazebo/TeleKyb3

MSc in Automation and Control Engineering

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Abstract — This report presents the design and simulation validation of an aerial pick-and-place mission performed by a fully actuated hexarotor equipped with a 3-DoF robotic manipulator and a two-finger gripper. The task consists of grasping a small cube from a pedestal, transporting it in flight, and releasing it onto a second pedestal. The proposed solution combines the TeleKyb3/GenoM control framework, state estimation through POM, trajectory generation with Maneuver, flight stabilization with NHFC, physical interaction handling through Phynt, and an online inverse-kinematics-based manipulation strategy.

Keywords — aerial manipulation; fully actuated hexarotor; pick-and-place; TeleKyb3; Gazebo; inverse kinematics; admittance filtering.

1. Introduction

Aerial manipulation extends the capabilities of flying vehicles from navigation and sensing to physical interaction with the environment [5, 6]. This is relevant for tasks such as inspection, light object transportation, and operations in areas that are difficult or unsafe to access with ground robots. At the same time, manipulation from an aerial platform is significantly more challenging than manipulation with a fixed-base robot, since the vehicle itself is part of the dynamic system and cannot be treated as an immovable support for the arm.

The objective of this project is to design and validate in simulation a pick-and-place mission using a fully actuated hexarotor [2] equipped with a 3-DoF robotic manipulator and a two-finger gripper. The task consists of grasping a small cube from a pedestal, transporting it in flight, and releasing it onto a second pedestal. The system is implemented in Gazebo using the TeleKyb3/GenoM framework, which provides the components for state estimation, trajectory generation, flight control, physical interaction handling, and low-level actuation of the robotic platform.

The main challenge is the coupling between the aerial vehicle, the manipulator, the gripper, and the object. Small errors in the platform position can be significant with respect to the cube size, while fast arm motions may disturb hovering stability. Moreover, contact with the cube or the pedestal must be carefully handled, since an

inaccurate approach can push the object away or destabilize the grasp. For this reason, the final strategy relies on top-down grasping and placing, continuous correction of the arm configuration using the object pose, and immediate gripper actuation when the cube enters the grasping or placing volume.

2. Problem Formulation

The problem is formulated at the level of the mission logic rather than as a complete dynamic control problem. The low-level stabilization of the vehicle is handled by the TeleKyb3/GenoM control components, while the mission script defines the reference positions, computes the manipulator inverse kinematics, monitors the cube pose, and triggers the gripper actions. For this reason, the formulation focuses on the platform, the mission geometry, and the conditions that determine whether grasping and placing are successful.

2.1. Platform Design

The robotic platform used in this project is a fully actuated hexarotor equipped with a 3-DoF serial manipulator and a two-finger gripper. The aerial vehicle provides the mobile base for the manipulation task, while the arm performs the fine motion required to reach the object during grasping and placing.

The fully actuated nature of the platform means that the vehicle can generate a six- dimensional wrench, composed of a force vector and a moment vector, through the coordinated action of its rotors. Following the notation used in the Aerial Robotics course notes [5], this relation can be expressed as

$$\mathbf{w}(\mathbf{u}_\lambda) = \begin{bmatrix} \mathbf{f}(\mathbf{u}_\lambda) \\ \mathbf{m}(\mathbf{u}_\lambda) \end{bmatrix} = \mathbf{F}\mathbf{u}_\lambda, \tag{1}$$

where $\mathbf{w} \in R^6$ is the wrench applied to the vehicle body, $\mathbf{f} \in R^3$ is the force, $\mathbf{m} \in R^3$ is the moment, $\mathbf{u}_\lambda$ collects the rotor thrust inputs, and $\mathbf{F}$ is the Full Allocation Matrix (FAM). The wrench generated by the rotors is expressed in the body frame attached to the center of mass of the hexarotor.

In a fully actuated configuration,

$$\operatorname{rank}(\mathbf{F}) = 6, \tag{2}$$

which means that the platform can locally vary the three force components independently from the three moment components. The thrust inputs are related to the thrust intensities generated by the propellers, which in turn depend on the rotor spinning velocities.

Although the platform is mechanically fully actuated, the adopted flight-control setup does not explicitly exploit this property to command tilt-free lateral motion. The mission script provides position and yaw references to the TeleKyb3 stack, and the vehicle is controlled through NHFC in `tilt_prioritized` mode. Therefore, in this project the hexarotor is mainly used as a stabilized aerial base for manipulation, while the full-actuation property remains a feature of the mechanical platform rather than a capability explicitly used by the mission-level controller.

At mission level, the vehicle is treated as the controlled aerial base. The complete coupled dynamics of the hexarotor, manipulator, gripper, and object are not explicitly modeled in the Python script. Instead, the low-level flight controller stabilizes the vehicle, while the mission logic limits the aggressiveness of the arm motion and uses dedicated gains during manipulation. The wrench observer and admittance filtering provided by Phynt are used to improve the handling of physical interactions and external disturbances during the pick-and-place phases.

The manipulator is modeled as a 3-DoF arm with one yaw joint and two pitch joints. Its configuration is denoted by the three revolute joint angles

$$\mathbf{q} = \begin{bmatrix} q_1 & q_2 & q_3 \end{bmatrix}^T, \tag{3}$$

where q_1 orients the arm in the horizontal plane, while q_2 and q_3 determine the vertical motion of the two links.

The arm kinematics used in the mission script follows standard serial-robot modeling conventions [1, 3] and is expressed in the body frame of the hexarotor. Denoting by $L_2 = 0.15\,\mathrm{m}$ and $L_3 = 0.20\,\mathrm{m}$ the effective link lengths used in the inverse kinematics, and by $z_s = -0.15\,\mathrm{m}$ the vertical offset of the shoulder pitch joint, the end-effector position is computed as

$$\begin{aligned} {}^B x_E &= \rho(\mathbf{q}) \cos q_1, \\ {}^B y_E &= \rho(\mathbf{q}) \sin q_1, \\ {}^B z_E &= z_s - L_2 \sin q_2 - L_3 \sin(q_2 + q_3), \end{aligned} \tag{4}$$

where $\rho(\mathbf{q}) = L_2 \cos q_2 + L_3 \cos(q_2 + q_3)$.
The orientation of the end-effector frame with respect to the body frame is approximated as

$$ {}^{B}\mathbf{R}_{E} = \mathbf{R}_z(q_1)\mathbf{R}_y(q_2 + q_3), \tag{5} $$

so that the local x axis is aligned with the last arm link, while the local y axis corresponds to the opening direction of the two gripper fingers. This frame is used to evaluate whether the cube is inside the gripper volume. Given the cube position expressed in the body frame, the cube position in the end-effector frame is

$$ {}^{E}\mathbf{p}_C = ({}^{B}\mathbf{R}_{E})^T \left({}^{B}\mathbf{p}_C - {}^{B}\mathbf{p}_E \right). \tag{6} $$

The gripper is closed when the cube lies inside the capture volume

$$ \begin{aligned} |{}^{E}x_C| &\le 0.060\,\mathrm{m}, \\ |{}^{E}y_C| &\le 0.055\,\mathrm{m}, \\ |{}^{E}z_C| &\le 0.045\,\mathrm{m}. \end{aligned} \tag{7} $$

The two fingers are modeled as prismatic joints moving along the local gripper y direction. At mission level, the gripper is therefore treated as a binary actuator: open during approach and release, closed during grasping and transfer.
The simulated platform is shown in Figure 1.

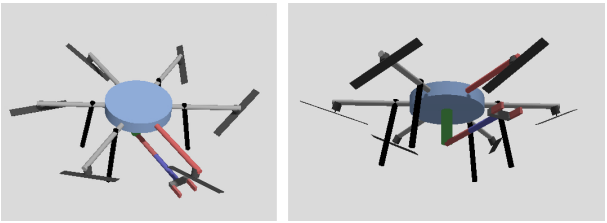


Figure 1: Gazebo model of the fully actuated hexarotor equipped with the 3-DoF manipulator and two-finger gripper used in the pick-and-place mission.

2.2. Mission Description

The objective of the mission is to move a cube from an initial pedestal to a second pedestal using only the aerial manipulator. The Gazebo world contains a first support table, a cube, and a second pedestal. The cube has side length 0.05 m and mass 0.05 kg, and its nominal initial center position is

$$ {}^{W}\mathbf{p}_{c,0} = \begin{bmatrix} 1.08 & 0 & 0.225 \end{bmatrix}^T \mathrm{m}. \tag{8} $$

The second pedestal is centered at

$$ {}^{W}\mathbf{p}_{p,2} = \begin{bmatrix} -1.00 & 0 & 0.20 \end{bmatrix}^T \mathrm{m}. \tag{9} $$

Both supports have a square top surface of approximately $0.20\,\mathrm{m} \times 0.20\,\mathrm{m}$, and the top surface is located at $z = 0.20\,\mathrm{m}$. Therefore, when the cube is correctly resting on a pedestal, its center is expected to be approximately at $z = 0.225\,\mathrm{m}$.
The mission is divided into a sequence of phases. First, the vehicle takes off with the manipulator stretched and reaches a cruise altitude of 1.00 m. It then moves close to the first pedestal and descends to the pick working altitude. The nominal horizontal position for grasping is computed from the current cube position as

$$ x_{\mathrm{pick}} = x_c - r_{\mathrm{pick}}, \qquad y_{\mathrm{pick}} = y_c, \tag{10} $$

where $r_{\mathrm{pick}} = 0.27\,\mathrm{m}$ is the nominal arm reach used in the implementation. Once the vehicle is close to this pose, the arm performs the fine alignment with the cube using inverse kinematics updated from the measured cube pose.
After a successful grasp, the vehicle lifts the cube and transfers it to the second pedestal. The placing phase is defined symmetrically with respect to the grasping phase, but with the vehicle yaw rotated by π so that the manipulator faces the target pedestal. The nominal place position is

$$ x_{\mathrm{place}} = x_{p,2} + r_{\mathrm{place}}, \qquad y_{\mathrm{place}} = y_{p,2}, \tag{11} $$

with $r_{\mathrm{place}} = 0.29\,\mathrm{m}$. The vehicle descends to a place working altitude of 0.68 m, while the gripper remains closed and the arm continuously corrects the cube position above the target pedestal.
The key mission parameters are summarized in Table 1.

Table 1: Main geometric parameters used in the mission formulation.

Parameter	Value
Initial cube center	(1.08, 0, 0.225) m
Second pedestal center	(−1.00, 0, 0.20) m
Cube side length	0.05 m
Cube mass	0.05 kg
Cruise altitude	1.00 m
Pick working altitude	0.62 m
Place working altitude	0.68 m
Pick reach offset	0.27 m
Place reach offset	0.29 m

The mission is considered successful if the cube is grasped from the first pedestal, transported without being dropped, and released on the second pedestal. The main constraints are the limited workspace of the manipulator, the need to avoid undesired contacts with the cube and the pedestals, and the requirement to keep the aerial platform sufficiently stable during grasping and placing. The detailed grasping, verification, recovery, and release logic is described in Section 4.

3. Simulation Setup

The proposed mission was developed and tested in a Gazebo simulation environment integrated with the TeleKyb3/GenoM framework. The simulation includes the platform, the 3-DoF manipulator, the two-finger gripper, the cube to be transported, and the two pedestals. The goal of the setup is to reproduce the main physical effects relevant to the task.

3.1. Gazebo World

In the adopted Gazebo setup, the physics engine uses a maximum step size of 0.002 s and a nominal real-time update rate of 500 Hz. Gravity is set to 9.81 m/s^2 along the negative vertical direction. The simulated scene contains the aerial platform, a first support table, a cube, and a second pedestal used as the target location for the release phase.
A simulated OptiTrack plugin publishes the pose of the hexarotor base at 100 Hz. This measurement is used by the TeleKyb3 state-estimation pipeline, reproducing the role of an external motion-capture system in an indoor aerial robotics experiment.

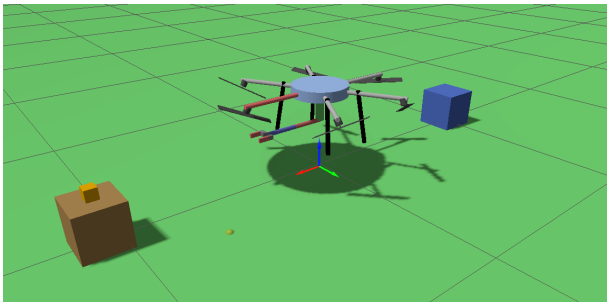


Figure 2: Gazebo simulation environment. The colored axes shown at the origin represent the world frame. The cube is initially placed on the first support, while the second pedestal defines the target location for the release phase.

3.2. Robot and Object Models

The aerial robot is modeled as a fully actuated hexarotor equipped with the serial manipulator described in Section 2.1. The arm joints are revolute joints controlled through the `dynamixel` component, while the gripper fingers are modeled separately as prismatic joints in Gazebo.
The cube is modeled as a solid box with side length 0.05 m and mass 0.05 kg. Its inertia is chosen consistently with a uniform solid cube. The cube collision surface uses increased friction coefficients in order to reduce slipping during grasping and transport. The same idea is applied to the gripper fingers, whose collision surfaces are also configured with high friction.
The gripper is composed of two fingers moving along opposite directions of the local gripper y axis. Each finger is connected to the gripper base through a prismatic joint with range from −0.02 m to 0.03 m. In the SDF model, these joints are controlled by Gazebo joint position controllers. For this reason, the mission script commands the gripper directly through Gazebo topics, instead of using the `dynamixel` component. At mission level, the gripper is treated as a binary actuator: an open command is used during approach and release, while a closed command is repeatedly sent during grasping and transfer to maintain contact with the cube.
Several physical parameters of the manipulator model are relevant for the simulation. The revolute arm joints include damping, which reduces impulsive reactions on the aerial platform dur-

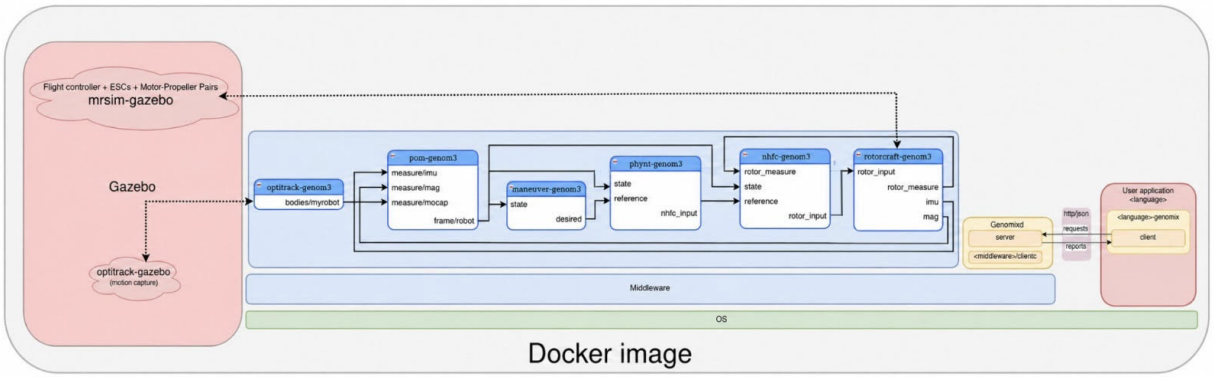


Figure 3: Overall simulation and control architecture used for the pick-and-place mission. The complete framework runs inside the Docker environment provided for the course. Gazebo simulates the robot, the object, and the environment, while the TeleKyb3/GenoM components handle state estimation, trajectory generation, physical interaction filtering, flight control, rotor actuation, and manipulator actuation.

ing arm motion. The finger joints have finite effort and velocity limits, so the gripper does not behave as an ideal rigid constraint. This is important because the cube can still slip or be released if the contact configuration is not sufficiently stable.

3.3. TeleKyb3/GenoM Framework

The simulation was executed inside the Docker environment provided for the course, using the TeleKyb3/GenoM framework documentation as reference [4]. This container includes the required Gazebo, TeleKyb3/GenoM, OpenRobots, and Python dependencies, allowing the simulation and control framework to run in a reproducible configuration. The overall software architecture is shown in Figure 3.

The mission is controlled through a set of GenoM components connected from the Python script. The simulated vehicle and its low-level sensors are interfaced through `rotorcraft`, while the pose of the hexarotor base is provided by a simulated motion-capture system through the Gazebo OptiTrack plugin and the GenoM `optitrack` component. The `pom` component estimates the vehicle state, `maneuver` generates smooth reference trajectories, `phynt` applies wrench observation and admittance filtering, and `nhfc` tracks the filtered reference by generating rotor commands. The manipulator joints are commanded through `dynamixel`, while the gripper fingers are commanded directly through Gazebo joint-position topics.

The main control chain is

$$\texttt{maneuver} \rightarrow \texttt{phynt} \rightarrow \texttt{nhfc} \rightarrow \texttt{rotorcraft}.$$

The `phynt` component receives the nominal reference from `maneuver`, the estimated vehicle state from `pom`, and the wrench measurement from `nhfc`. It outputs the filtered reference `phynt/desired`, which is then tracked by `nhfc`. This structure allows the nominal waypoint trajectory to be modified by the admittance layer before being sent to the flight controller.

The `nhfc` component is configured with the mass, rotor geometry, thrust and torque coefficients, rotor limits, control gains, and the `tilt_prioritized` attitude control mode. Its output is the rotor command sent to `rotorcraft`. The arm and gripper are handled outside this flight-control chain: the revolute joints of the arm are controlled through `dynamixel`, while the gripper is commanded through Gazebo topics because its fingers are modeled as prismatic joints that are not included in the `dynamixel` component.

4. Control and Manipulation Strategy

This section describes the control and manipulation strategy implemented to make the pick-and-place task robust in simulation. The main idea is to use the aerial vehicle for coarse positioning and stable hovering, while the manipulator performs the fine motion required to grasp

and release the cube. Therefore, the mission logic combines conservative flight references, interaction handling through `phynt`, online inverse kinematics, geometric grasping and release conditions, and explicit verification steps.

4.1. Interaction Handling and Stable Hovering

The aerial vehicle is not required to place the gripper exactly around the cube by flight motion alone. Instead, the drone is commanded to reach a useful standoff position with respect to the cube or to the target pedestal. Once this coarse positioning is achieved, the vehicle should remain as still as possible, while the arm compensates for the remaining error.

For this reason, different trajectory profiles are used depending on the mission phase. Cruise motions can be faster, while take-off, approach, carry, lift, and landing phases use smoother limits on velocity, acceleration, jerk, and snap. This reduces aggressive transients and limits the oscillations of the aerial base when the arm is moving or when the cube is held by the gripper. If a command sent to `maneuver` is rejected due to dynamic limits, the same target is retried with a longer duration or with a recovery profile. The dynamic limits were selected empirically from simulation tests. For example, the transfer between the two pedestals uses a particularly conservative profile to reduce the risk of losing the cube during flight.

The flight-control gains are also changed during manipulation. In pick and place phases, a more damped set of `nhfc` gains is used. The horizontal proportional and derivative actions are increased to reduce residual motion, while the horizontal integral gain is reduced. This prevents excessive integral accumulation when the arm introduces slowly varying disturbances. After manipulation, the nominal flight gains are restored for transfer and return flight.

Before enabling the admittance filter, consistently with the wrench-observer and admittance-control concepts discussed in the course material [5], the wrench observer bias is reset while the vehicle is hovering in the manipulation configuration. This makes the current hover condition the nominal equilibrium of the interaction filter. Constant offsets due to the arm configuration, model mismatch, or controller bias are absorbed in the observer bias, while variations of the observed wrench during arm motion or contact are treated as interaction disturbances. In this way, `phynt` improves interaction handling without introducing a permanent displacement of the desired hovering position.

4.2. Online Arm Servoing and Inverse Kinematics

Once the drone has reached a useful standoff position, the remaining alignment error is handled by the manipulator. The arm motion is not planned as a fixed trajectory. Instead, the target pose is continuously updated from the measured cube pose and from the estimated drone pose. This is important because the aerial base is never perfectly stationary, and even small position errors are significant with respect to the cube size.

The target point is first defined in the world frame. During grasping, this target is obtained from the current cube position plus a small offset used to approach the cube from above. During placing, the target is related to the desired release position above the second pedestal. At each update, the target is converted into the drone body frame using the current estimated drone position and yaw:

$$ {}^{B}\mathbf{p}_T = \mathbf{R}_z(-\psi_D)\left({}^{W}\mathbf{p}_T - {}^{W}\mathbf{p}_B\right), \tag{12} $$

where ${}^{W}\mathbf{p}_T$ is the target position in the world frame, ${}^{W}\mathbf{p}_B$ is the drone position, and ψ_D is the drone yaw angle estimated by `pom`.

The inverse kinematics is derived from the same convention introduced in Section 2.1 and from standard geometric IK arguments [3]. Given ${}^{B}\mathbf{p}_T = [{}^{B}x_T, {}^{B}y_T, {}^{B}z_T]^T$, the yaw joint is computed as

$$ q_1 = \operatorname{atan2}({}^{B}y_T, {}^{B}x_T). \tag{13} $$

The remaining problem is planar. Defining

$$ R_T = \sqrt{({}^{B}x_T)^2 + ({}^{B}y_T)^2}, \qquad h_T = {}^{B}z_T - z_s, \tag{14} $$

and

$$ D_T = \sqrt{R_T^2 + h_T^2}, \tag{15} $$

the elbow angle is obtained from the law of

cosines:

$$\cos\theta_3 = \frac{D_T^2 - L_2^2 - L_3^2}{2L_2L_3}. \tag{16}$$

The corresponding shoulder angle in the planar geometric convention is:

$$\begin{aligned}\theta_2 = &\operatorname{atan2}(h_T, R_T) \\ &- \operatorname{atan2}\left(L_3\sin\theta_3, L_2 + L_3\cos\theta_3\right).\end{aligned} \tag{17}$$

Since the Gazebo joint convention used in the model is the one adopted in Eq. (4), the planar solution is mapped to the commanded joint angles as:

$$q_2 = -\theta_2, \qquad q_3 = -\theta_3. \tag{18}$$

Both elbow configurations can be evaluated, and solutions outside the joint limits are discarded. During top-down grasping and placing, the selected branch is constrained so that the arm approaches the object from above rather than with a lateral scoop-like motion.
The IK computation is repeated online during the manipulation phases. The cube pose is read from Gazebo, while the drone pose is read from the `pom` estimate. Both signals are filtered before being used in the IK loop:

$$\hat{\mathbf{p}}_k = (1-\alpha)\hat{\mathbf{p}}_{k-1} + \alpha\mathbf{p}_k. \tag{19}$$

The filtered target reduces the effect of pose jitter, while still allowing the arm to react quickly to cube motion. The resulting joint reference is also slew-rate limited before being sent to the joint controller, so that the target evolves smoothly even when the IK solution changes abruptly.
The revolute joints are actuated through `dynamixel` using a velocity-level proportional controller:

$$\dot{q}_{i,\mathrm{cmd}} = \mathrm{sat}_{\dot{q}_{\max}}\left(K_p(q_{i,\mathrm{ref}} - q_i)\right). \tag{20}$$

The commanded velocity is additionally acceleration-limited. This prevents impulsive arm motions, which would disturb the hovering vehicle and could destabilize the grasp. Therefore, the arm behaves as a continuous servoing system: the IK defines the desired configuration, while the velocity controller makes the joints converge smoothly to it.

4.3. Grasping Logic and Verification

The grasping strategy uses a top-down, or slightly oblique top-down, approach. The drone is first positioned behind the cube:

$$x_{\mathrm{pick}} = x_c - r_{\mathrm{pick}}, \qquad y_{\mathrm{pick}} = y_c. \tag{21}$$

The arm then moves toward a point above and slightly behind the cube, and only after this approach does it move closer to the top region of the cube. This avoids a lateral "scoop" motion, which could push the cube away from the pedestal before the gripper closes.
The gripper is closed as soon as the cube enters the useful volume of the fingers, defined in Eq. (7). This local condition is more reliable than checking only the distance between the cube and the end-effector in world coordinates, because it verifies that the cube is actually between the fingers.
Once the condition is satisfied, the gripper command is sent immediately and the drone climbs to a safer altitude. The grasp is then verified by reading the cube height. If the cube is above the pedestal height, the grasp is confirmed. If the cube is still close to the pedestal, the grasp is rejected and a new attempt is performed. In that case, the gripper is opened and the arm is stretched while the drone is still high, before descending again. If the cube is detected below the pedestal height, the mission is aborted because the object has fallen.

4.4. Transfer and Placing Strategy

After a confirmed grasp, the vehicle transfers the cube to the second pedestal with the arm stretched and the gripper kept closed. Keeping the arm stretched reduces the motion of the load with respect to the vehicle and avoids carrying the cube close to the pedestal or to the vehicle body. During this phase, the gripper close command is also periodically refreshed, so that the simulated finger controllers keep applying a closing action throughout the transfer.
A specific modification was introduced to improve the stability of the grasp. In the SDF model, the maximum effort of the prismatic joints actuating the gripper fingers was increased from 25 N to 100 N. This does not modify the high-level mission logic, but it increases the maximum force that the simulated finger joint

controllers can apply when the gripper is commanded to close. Together with the repeated close commands, this reduced the probability of losing the cube during flight, especially during the transfer between the two pedestals.
The placing phase follows the same principle used for grasping. The vehicle performs a coarse positioning motion near the second pedestal, while the manipulator performs the fine correction. Since the vehicle is rotated by π during placing, the nominal placing position is chosen as

$$x_{\text{place}} = x_{p,2} + r_{\text{place}}, \qquad y_{\text{place}} = y_{p,2}. \tag{22}$$

The drone then descends to a placing altitude that is high enough to reduce the risk of collisions with the pedestal, the cube, or the landing legs. From this configuration, the arm continuously updates its inverse-kinematics target in order to move the held cube above the center of the second pedestal.
The release condition is evaluated on the measured cube pose, not only on the end-effector pose. This is important because the cube is not rigidly attached to the gripper in the simulation, and its position can differ slightly from the nominal end-effector position. The gripper is opened only if the cube lies inside a release volume centered above the second pedestal:

$$\sqrt{(x_c - x_{p,2})^2 + (y_c - y_{p,2})^2} \leq 0.085\,\text{m}, \tag{23}$$

$$z_{\min} \leq z_c \leq z_{\max}. \tag{24}$$

In the implementation, these bounds are chosen around the desired release height z_{rel}, defined as the pedestal height plus half of the cube side length and a small clearance:

$$z_{\text{rel}} = z_{\text{ped}} + \frac{l_c}{2} + \Delta z_{\text{rel}}. \tag{25}$$

In the final tuning, the clearance was set to

$$\Delta z_{\text{rel}} = 0.018\,\text{m}, \tag{26}$$

which gives

$$z_{\text{rel}} = 0.243\,\text{m}. \tag{27}$$

The release interval is then set to

$$z_{\min} = z_{\text{rel}} - 0.012\,\text{m}, \qquad z_{\max} = z_{\text{rel}} + 0.014\,\text{m}. \tag{28}$$

If this condition is not satisfied, the gripper remains closed and the arm continues to correct the cube position. When necessary, the drone standoff is also adjusted to bring the release target back inside the arm workspace. Once the cube enters the release volume, the gripper is opened and the vehicle climbs before stretching the arm and returning to the landing area.

4.5. Implementation Details

Some additional implementation details were introduced to make the control loop more reliable in simulation. First, the cube pose is read from Gazebo in a background thread and stored in a short-lived cache. This avoids blocking the online IK loop with repeated `gz topic` calls, while still providing an updated cube position during grasping and placing. A similar caching strategy is used for the drone pose estimated by `pom`, which is read periodically and averaged over recent samples.
Gripper commands are also sent asynchronously through Gazebo topics. This was necessary because waiting for each topic publication to terminate could delay the grasping logic. For this reason, open and close commands are queued without blocking the mission script, and the close command is repeated during grasping and transfer.
Finally, small offsets are used when retrying a failed grasp while the cube is still on the pedestal. Instead of repeating exactly the same target, the following attempts scan a small neighborhood around the measured cube position. During placing, the controller also estimates the current offset between the end-effector and the held cube, so that the measured cube position, rather than only the nominal end-effector pose, is driven toward the release volume.

5. Results and Discussion

This section presents the results obtained with the final implementation and discusses the main behaviors observed during simulation. The analysis focuses on the picking phase, which is the most critical part of the mission, and on the role of the interaction filtering provided by `phynt`. The results are used to evaluate whether the vehicle can maintain a sufficiently stable hovering condition while the manipulator performs

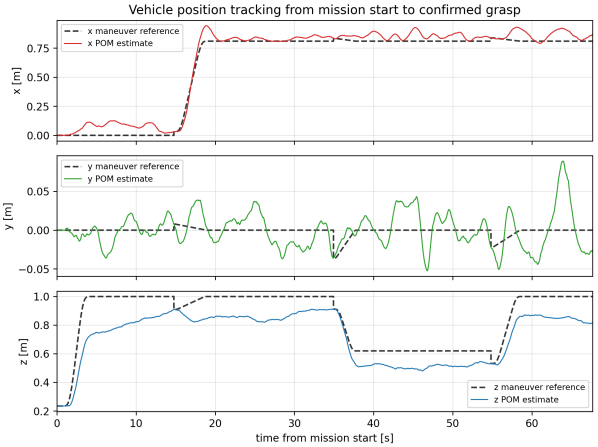


Figure 4: Vehicle position tracking during the picking phase. The dashed curves are the reference generated by `maneuver`, while the solid curves are the platform position estimated by `pom`.

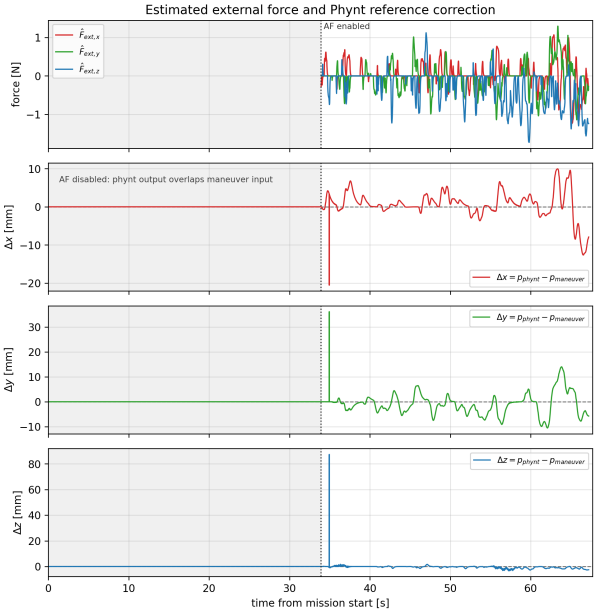


Figure 5: Estimated external force and position correction introduced by `phynt` during the picking phase. The lower plots show $\mathbf{p}_{\texttt{phynt}} - \mathbf{p}_{\texttt{maneuver}}$ in millimeters.

the fine motion required to grasp the cube.

5.1. Vehicle Tracking During Picking

Figure 4 shows the position tracking of the aerial platform from the beginning of the mission to the confirmed grasp. The solid lines represent the vehicle position estimated by `pom`, while the dashed lines represent the reference generated by `maneuver`. This reference is not the high-level waypoint commanded by the mission script, but the time-parametrized trajectory generated by the `maneuver` component and sent to the rest of the control chain.

For this reason, small discontinuities can be observed when a new maneuver starts. At those instants, the trajectory generator is reinitialized from the current estimated vehicle state and then evolves toward the next commanded waypoint. These transitions are visible, for example, during the motion toward the picking area and during the descent to the manipulation altitude. The horizontal tracking is sufficiently accurate for the manipulation strategy adopted in the mission. The vehicle reaches the neighborhood of the first pedestal and remains close to the desired horizontal position while the manipulator performs the fine alignment with the cube. This behavior is consistent with the design choice of using the drone mainly for coarse positioning and relying on the arm for the final grasping motion.

The vertical tracking error is larger than the horizontal one, especially during the picking phase. In some parts of the maneuver, the difference between the estimated vertical position and the generated reference is of the order of 0.1–0.2 m. This residual offset is also consistent with the hover diagnostics observed during the simulation. It can be attributed to model mismatch associated with the manipulator and gripper, the conservative controller tuning used during manipulation, and the fact that the comparison is made with the generated `maneuver` reference rather than with an ideal piecewise-constant mission waypoint. Despite this vertical error, the platform remains stable enough for the arm to complete the top-down grasping procedure.

5.2. Effect of Phynt Admittance Filtering

Figure 5 shows the external force estimated by the wrench observer and the position correction introduced by `phynt` during the picking phase. The correction is computed as

$$\Delta\mathbf{p} = \mathbf{p}_{\texttt{phynt}} - \mathbf{p}_{\texttt{maneuver}}, \tag{29}$$

where $\mathbf{p}_{\texttt{maneuver}}$ is the nominal reference generated by `maneuver`, and $\mathbf{p}_{\texttt{phynt}}$ is the filtered reference sent to `nhfc`.

Before enabling the admittance filter, `phynt` is expected to behave as a pass-through block. The available pre-filter samples confirm that the difference between `phynt/desired` and `maneuver/desired` is small, of the order of a few millimeters. In the analyzed log, the earliest part of the plot is represented as zero correction because no `phynt` samples were available for that interval.
The correction remains small for most of the picking phase. This is expected from the chosen admittance parameters. In a simplified quasi-static interpretation,

$$\Delta p_i \simeq \frac{\hat{F}_{ext,i}}{K_i}, \tag{30}$$

where K_i is the virtual stiffness along direction i. Since the implemented stiffness values are

$$K_x = K_y = 50\,\mathrm{N/m}, \qquad K_z = 500\,\mathrm{N/m},$$

forces of the order of 0.1–0.5 N generate corrections of only a few millimeters. For example,

$$0.2\,\mathrm{N}/50\,\mathrm{N/m} = 4\,\mathrm{mm}, \qquad 0.5\,\mathrm{N}/500\,\mathrm{N/m} = 1\,\mathrm{mm}.$$

This explains why the effect of `phynt` is visible only when plotting the difference directly.
A larger peak appears shortly after the filter is enabled. This peak is mainly due to the reinitialization of a new `maneuver` trajectory during the descent toward the picking altitude, not to a steady-state admittance displacement. After this short transient, the correction returns to a small value.
Overall, the plot confirms that `phynt` is active, but also that its tuning is intentionally stiff. In this project, the admittance filter is used to improve interaction handling without introducing large motion of the aerial base during grasping.

5.3. Mission Outcome

The robustness of the final implementation was evaluated over ten simulation runs. The outcome of the main mission phases is summarized in Table 2. A grasp at first attempt indicates that the cube was successfully captured without requiring a retry, while a successful grasp means that the cube was eventually lifted above the first pedestal. A successful transfer requires the cube to remain in the gripper during the flight toward the second pedestal, and a successful place is counted when the cube is released on the target pedestal.

Table 2: Mission success statistics over ten simulation runs.

Metric	Runs
Grasp at first attempt	5/10
Successful grasp	8/10
Successful transfer	7/10
Successful place	6/10
Complete mission success	6/10

6. Conclusions

This project implemented a complete aerial pick-and-place mission in Gazebo using a fully actuated hexarotor equipped with a 3-DoF manipulator and a two-finger gripper. The final strategy combines coarse vehicle positioning, online arm inverse kinematics, top-down grasping and placing, gripper-volume checks, and grasp verification after lifting.
The mission was successfully completed in simulation, showing that the adopted TeleKyb3/GenoM architecture can coordinate flight control, manipulation, and interaction handling through `phynt`. The main limitations are the sensitivity to vertical tracking errors, contact dynamics, and the exact release configuration above the second pedestal. The solution also relies on the cube pose provided by Gazebo, so perception uncertainty is not considered. For this reason, the results should be interpreted as a validated simulation outcome rather than as a controller immediately transferable to hardware without further tuning. Future work should focus on perception, model calibration, and validation on real hardware.

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